



Understanding the diffusion of AI-generative (ChatGPT) in higher education: Does students' integrity matter?

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ABSTRACT

ChatGPT, an AI-powered language model, is revolutionising the academic world. Scholars, researchers, and students use its advanced capabilities to achieve their educational objectives, including generating innovative ideas, delivering assignments, and conducting extensive research projects. Nevertheless, the use of ChatGPT among students is contentious, giving rise to significant apprehensions regarding integrity and AI-facilitated deceit. At the same time, scholarly communities currently need more well-defined standards for adopting such academia-oriented technology. This study aims to determine students' use of ChatGPT using the Unified Theory of Acceptance and Use of Technology (UTAUT) and Social Cognitive Theory (SCT), notably the role of students' integrity in determining adoption behaviour. The analysis of 921 responses demonstrated that the utilisation of ChatGPT is influenced positively by performance expectancy, social influence, educational self-efficacy, technology self-efficacy, and personal anxiety. Conversely, student integrity was found to negatively impact usage. Remarkably, student integrity has a positive moderating effect between effort expectancy and ChatGPT usage. At the same time, it has a negative moderating effect on the link between performance expectancy and technology self-efficacy with ChatGPT usage. Hence, we propose that the academic community, AI language model developers, publishers, and relevant stakeholders collaborate to establish explicit rules for the utilisation of AI chatbots in an ethical manner for educational purposes.

1. Introduction

Recently, the world has been experiencing unprecedented advancements in modern technology as Artificial Intelligence (AI) keeps growing and advancing, sparking intense interest across industries (Sætra, 2023). With a plethora of innovations, intelligent Chatbot tools GPT (Generative Pre-trained Transformer) demonstrate how significantly AI has

progressed. ChatGPT, an AI-powered processing tool, was launched by OpenAI in November 2022. It quickly emerged as among the most critical and groundbreaking AI platforms (Forbes, 2023; Open AI, 2023). Chatbots generate writing which mimics that of humans via a conversational procedure. These chatbots can handle various language demands such as question-answering, translation, summarisation, and text production (Cotton et al., 2023). Exceptionally, ChatGPT is experiencing

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phenomenal growth as a user application on the internet. According to Bin-Nashwan, Ismaiel, et al. (2023), it has amassed almost 100 million users as of January 2023. Additionally, it attracts approximately 1.8 billion website views per month, as Forbes (2023) and Reuters (2023) reported. ChatGPT has been employed for various applications, including content generation (Bloomberg, 2023; Reuters, 2023), where it demonstrated the ability to produce articles (Benichou, 2023; Lo, 2023), facilitate the learning process (Hadi Mogavi et al., 2023), and generate other types of authentic and logical written content.

ChatGPT has proven to be a double-edged sword, causing ripples in the academic community at the back of its extraordinary success (Benichou, 2023; Bin-Nashwan, Ismaiel, et al., 2023; Hadi Mogavi et al., 2023; Strzelecki & ElArabawy, 2024). Some potential uses for such a revolutionary tool include for efficiency improvement (Lada, Chekima, Karim, et al., 2023; Rice et al., 2023), assisting academics and students in coming up with ideas and getting past writer's block (Cotton et al., 2023; Pfau et al., 2023; Rice et al., 2023) and serving as a tool for automating repetitive and time-consuming content generation (Bloomberg, 2023; Reuters, 2023). However, the academic usage of ChatGPT has its own set of challenges, including the tendency to commit academic fraud and plagiarism (Bin-Nashwan, Ismaiel, et al., 2023; Lo, 2023; Strzelecki, 2023). AI-enabled platforms for writing essays facilitate students and researchers in producing articles, research papers, and assignments, thus giving rise to the issues of cheating and academic integrity (Bin-Nashwan, Ismaiel, et al., 2023; Cotton et al., 2023). As such, on-campus access to ChatGPT and other AI tools has been banned in certain universities globally (Tlili et al., 2023) – whilst others remain unwilling to follow suit (Huang, 2023) – due to concerns of plagiarism and false contents. Meanwhile, instead of calling for a ban, experts on technology have advised universities to conduct training on ChatGPT and AI platform usage so that students, researchers, and faculties would use the tools properly (Jimenez, 2023). This clarifies the significance and applicability of our study. It is imperative that we uncover the motives for students' usage of ChatGPT and how their integrity plays a part in their decision to adopt this cutting-edge tool.

Despite the increasing number of studies examining the possible role, effects, prospects, and concerns with regard to ChatGPT utilisation in education (Benichou, 2023; Bin-Nashwan, Ismaiel, et al., 2023; Cotton et al., 2023; Hadi Mogavi et al., 2023; Pfau et al., 2023; Tlili et al., 2023), very few had explored the factors driving students to utilise ChatGPT specifically from the lens of the Unified Theory of Acceptance and Use of Technology (UTAUT) and Social Cognitive Theory (SCT), as well as the role of integrity in driving the students' adoption behaviour. In academia and higher education, students' learning integrity has consistently been a problem; with the widespread adoption of AI chatbots, this concern has become even more prominent (Dwivedi et al., 2023). Consequently, this present work attempts to close this gap by examining the variables that affect students' use of ChatGPT by adding academic integrity as a moderating factor to the research model. Bandura's (1989) SCT and Venkatesh et al.'s (2003) UTAUT model have been integrated to provide an all-inclusive and well-structured approach for examining the situational and personal factors affecting ChatGPT adoption among students. The current work intends to provide answers to these queries: 1) What are the relationships between students' use of ChatGPT and the factors of performance expectancy, effort expectancy, social influence, educational self-efficacy, technology self-efficacy, and personal anxiety? and 2) Does integrity moderate the aforementioned relationships in the ChatGPT adoption model? The results of this work have significant implications for the academic community's stakeholders in addition to enriching the existing body of knowledge.

2. Literature review

2.1. Theoretical foundation and research model

Individual behaviour has been broadly investigated in numerous

socio-cultural settings utilising prominent theoretical frameworks. Whilst prevailing frameworks have been employed, authenticated, revised, or panned, newer ones are progressing and being built to solve the faults/shortfalls in the former (Izuagbe et al., 2019; Ma & Huo, 2023). The original and variations of the TAM, TAM2 and TAM3, DOI, SCT, AIDUA and UTAUT are the most prominent theories/models for studying the predispositions to adopt, implement, and utilise technology. The TAM and UTAUT models are considered the most advantageous (Bouteraa, Raja Rizal Iskandar, & Zairani, 2022, 2023). With its high explanatory power and ability to predict, UTAUT has been determined to be the best for describing technology diffusion (Souiden et al., 2020; Strzelecki & ElArabawy, 2024).

In order to provide a thorough framework for comprehending users' behaviour in relation to technology, the UTAUT model integrates eight main models/theories, namely the TRA, the TAM, MM, TPB, the combination of TBP and TAM, the PC utilisation model, DOI, and SCT. Users' behaviour was acknowledged to be captured by performance expectancy, effort expectancy, social influence, and facilitating conditions. The UTAUT model outperformed the existing models by explaining 70% of the variance in technology acceptance. Since the function of ChatGPT is highly relevant to technological elements, it is rational to grasp its diffusion from the perspective of information systems research as suggested by Yilmaz and Yilmaz (2023a) and Yilmaz and Yilmaz (2023b). Thus, UTAUT is used in this work as the underlying basis for investigating students' use of ChatGPT (Strzelecki, 2023; Yilmaz et al., 2023).

According to Souide et al.'s (2020) systematic review, UTAUT has been proven to be reliable for analysing how people embrace new IT systems. However, the individual situations of users may limit the explanatory prowess of the model (Tarhini et al., 2016). Likewise, Venkatesh et al. (2012) recommended that the integration of additional related contributing factors could improve the prospects of the UTAUT in the users' context.

In this regard, the SCT theory by Bandura (1989) is also integrated as an underpinning theory to further strengthen the theoretical basis of the study's model. The basic tenet of SCT is that behavioural intentions are shaped by a person's actions as well as by environmental, cognitive, and personal factors. The core idea of the SCT is that people's behaviour is influenced cognitively as well as by circumstances in the external social environment. The theory of triadic reciprocal determinism, introduced by Bandura (1989), links physical characteristics, such as elements of the external environment, with personal characteristics, such as interior cognitive and affective emotions, all of which interact to influence overt behaviour. People act in certain ways based on their expectations, beliefs, and perceptions. Moreover, the idea suggests that an individual's aptitude, expertise, and understanding influence their motivation to carry out particular tasks.

Bandura (1989) highlighted further that the environment, which includes outside factors, shapes human behaviour. There are physical and sociological elements to such an ecosystem. The natural and man-made components of an individual's surroundings are included in the physical environment. Elizabeth and Michele (2001) proposed that the social environment consists of the physical surroundings, cultural settings, and social interactions of people. In fact, it includes elements like values, peer influence, community access, social standards, and more (Bandura, 1989). Behaviour, or how a person acts or responds in certain circumstances or towards particular objects, is the third component of the SCT (Bandura, 1991). This idea also encompasses a person's interaction with new developments in technology (Boateng et al., 2016). These three elements interact with one another to predict people's behaviour. However, their predictive capacities differ, and their mutual impacts do not always happen at the same time (Bandura, 1989).

As such, these essential aspects of the SCT and UTAUT as the underpinning theories can provide an integrated model to holistically comprehend ChatGPT usage among students, considering a variety of behavioural, personal, and environmental factors such as performance

expectancy, effort expectancy, social influence, educational self-efficacy, technology self-efficacy, and personal anxiety as influencing factors and students' integrity (Fig. 1).

2.2. Performance expectancy

Researchers have focused a lot of attention on performance expectancy as a way to forecast users' belief in task enhancement via technologies. Performance expectancy is considered the best predictor of behavioural intention based on UTAUT (Venkatesh et al., 2003) and UTAUT2 (Venkatesh et al., 2012). Performance expectancy has been indicated as the primary construct elucidating improved information system use (Azman Ong et al., 2023; Bouteraa et al., 2021; Tewari et al., 2023). Specifically, ChatGPT's usefulness can only be fully appreciated to the degree that it fulfils users' expectations (Sharma et al., 2022). The literature emphasised the cruciality of performance expectancy in influencing how users behave while utilising ChatGPT across a range of industries such as travel and tourism (Melián-González et al., 2021), healthcare (Shahsavari & Choudhury, 2023), and life coaching (Terblanche & Kidd, 2022). However, less consideration has been given to investigating its role in the academic setting. In keeping with the literature, if students perceive that ChatGPT would significantly improve their academic performance and provide various amenities such as efficacy, utility, ease, and apt tasks, they are likely to adopt its usage. The arguments above lead to the development of this hypothesis:

H1. Performance Expectancy positively affects students' use of ChatGPT.

2.3. Effort expectancy

According to the UTAUT model, effort expectancy significantly predicts people's propensity to use technology. It has to do with how easy people think technology is to use and how little mental or physical work they think it would need (Venkatesh et al., 2003). In short, individuals tend to embrace technology if they perceive it to be easy to use. Several studies revealed that effort expectancy significantly affects consumers' intentions (Bin-Nashwan, 2022; Bouteraa, Al-Daihani, et al., 2023; Rahim et al., 2022). Building on this, individuals' effort expectancy would be essential in determining whether they would use ChatGPT. Thus, students are likelier to adopt ChatGPT if they believe the platform requires little learning curve or time commitment (Menon & Shilpa, 2023). Accordingly, ChatGPT's convenience and ease of use can positively influence students' willingness to utilise it. The following is

hence hypothesised:

H2. Effort Expectancy positively affects students' use of ChatGPT.

2.4. Social influence

Social psychology theories promote social impact on behaviour. The conflict elaboration theories of social effect identified that adopting or rejecting an invention alters group interactions (Mugny et al., 1995). Kelman (1974) stated that compliance, internalisation, and identification are the three most basic forms of influence. Social influence indicates how much others' opinions affect technology use (Venkatesh et al., 2003), as proposed by various technology acceptance and use models, including the TRA, TBP, TA, and UTAUT. Technology-related literature indicates that others' inducements, preferences, and expectations strongly influence users' behaviour (Al-Saedi et al., 2020; Bin-Nashwan, Ismaiel, et al., 2023; Gunasinghe et al., 2020). Still, other investigations did not confirm this (Abbasi et al., 2021; Al-Hattami, 2023; Bouteraa, Raja Rizal Iskandar, & Zairani, 2022, pp. 64–102). Conflicting results show that peer influence is situation-dependent, motivating this study's context analysis. It could be construed that ChatGPT is a trendy and modernistic tool, lending credence to the concept that students' use of AI Chatbots is influenced by their peers. This leads to the hypothesis below:

H3. Social influence positively affects students' use of ChatGPT.

2.5. Education self-efficacy

The SCT developed by Bandura (1989) introduced the concept of self-efficacy, which explains an individual's confidence in his/her capability to perform or learn certain tasks. When used in the academic setting, it frequently relates to academic or education self-efficacy, which is characterised as assurance in reaching learning objectives and success in the classroom (Bong & Skaalvik, 2003) and is related to motivation, emotion, cognition, self-regulation, and achievement (Bong & Skaalvik, 2003). Following ChatGPT's release, academics have been engrossed and concerned; in the meantime, there are supporters and detractors of the tool's usage among researchers and students. Nonetheless, it is advantageous to draw conclusions and formulate qualified judgements regarding ChatGPT's importance in teaching (Rudolph et al., 2023). According to Cope et al. (2021), AI-powered platforms have great promise for revolutionising educational environments. Through the provision of access to a sophisticated and potent instrument

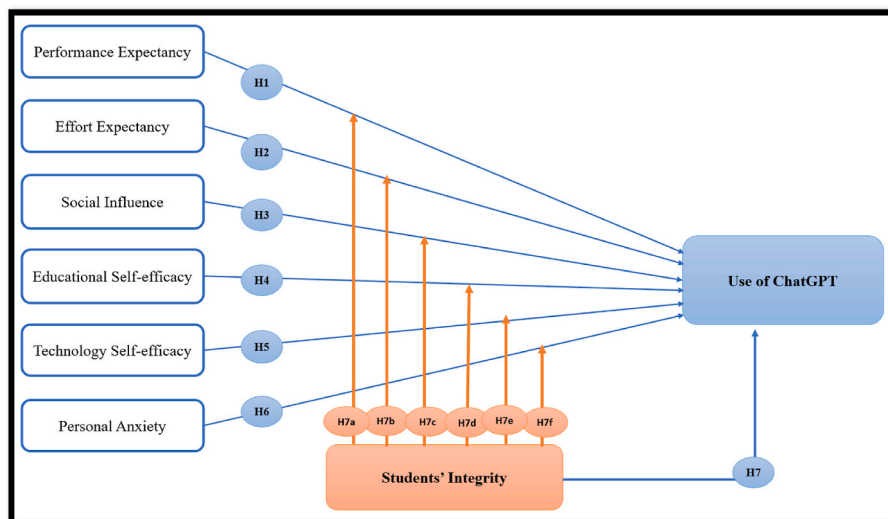


Fig. 1. Research model.

which expands the students' potential, AI usage in higher education can potentially increase academic self-efficacy (Jürgen Rudolph et al., 2023). Likewise, ChatGPT usage can lessen students' workloads, assess the material critically, and enhance their learning (Cotton et al., 2023; Tlili et al., 2023). Few research had looked into how academicians' and professors' use of ChatGPT is impacted by their sense of academic self-efficacy (Bin-Nashwan, Ismaiel, et al., 2023). Nevertheless, the influence of education self-efficacy on students' use of ChatGPT has not been thoroughly examined in many empirical investigations, necessitating this current study. It is formally hypothesised, in light of the prior claims that ChatGPT will help students grasp their material and get past obstacles in the learning process, that:

H4. Education self-efficacy positively affects students' use of ChatGPT.

2.6. Technology self-efficacy

This construct is a part of the SCT developed by Bandura (1989). It entails how a person perceives his capability to perform or accomplish specific tasks. Self-efficacy is subject- and situation-specific. Technology self-efficacy, therefore, refers to the confidence in achieving success and objectives using technology systems (Yoon et al., 2020). Self-regulation is related to an individual's drive, achievement, emotion, cognition, and self-control (Bong & Skaalvik, 2003). In the context of students' learning, technological self-efficacy crucially determines how well they can utilise technological tools for educational purposes. Following the release of ChatGPT, technology self-efficacy can influence students' use of ChatGPT or similar AI-powered tools in several ways; for instance, their belief in their ability to use technology effectively may cause them to be more open to incorporate ChatGPT into their learning processes and to explore its capabilities (Jürgen Rudolph et al., 2023). In addition, students with high confidence in their capability to use technology tend to participate in independent learning with tools like ChatGPT. They may find it as a resource for self-directed exploration and problem-solving (Cotton et al., 2023; Tlili et al., 2023). However, inadequate studies had empirically validated the impact of technology self-efficacy on the utilisation of ChatGPT among students, hence necessitating this investigation. Based on the earlier assertions, it can be stated as a formal hypothesis that:

H5. Technology Self-Efficacy positively affects students' use of ChatGPT.

2.7. Personal anxiety

Anxiety denotes the extent to which a person's apprehensions affect his anticipated or actual usage of a given system upon interacting with the technology (Verkijika, 2020). The mental accounting theory by Thaler (1980) shows that users' decision-making process is associated with transactional features of benefits, risks, and uncertainties. Anxiety — drawing upon Bandura's (1989) SCT — is a predictive behavioural factor which focuses on the psychological state of individuals in terms of competence and inclination to utilise technological tools (Meuter et al., 2003). The emergence of personal anxiety is triggered by the person's uncontrollable need to cope with continuous problems and hindrances related to key life or work matters.

Past studies have demonstrated the negative effect of anxiety on mental health in the context of e-learning (Almomani et al., 2021; Heo & Han, 2018). Stress levels were shown to spike during the COVID-19 pandemic, specifically due to the hasty and enforced move to online learning (Mheidly et al., 2020). Likewise, studies found a favourable correlation between positive views of technology and mental health (Nguyen & Chen, 2023), suggesting that ChatGPT's insight as an objective instrument is linked to user satisfaction and reduced anxiety. Abd-Alrazaq et al. (2022) carried out a systematic review to assess chatbot effectiveness. The results were inadequate and showed scant evidence that chatbots can aid in the management of stress, anxiety, and

depression. The use of chatbots had no discernible impact on the subjective psychological well-being. As a result, the research on the subject of chatbots and anxiety intensity was equivocal. In instances where students worry about finishing their homework or research tasks, ChatGPT can be a valuable tool to help explain ideas and provide clarifications, potentially lowering anxiety associated with academic assignments. However, due to the dearth of empirical evidence, especially in the context of academia, more research is required to validate this conclusion (Abd-Alrazaq et al., 2022). As such, it can be hypothesised that:

H6. Personal anxiety positively affects students' use of ChatGPT.

2.8. Students' integrity

Honesty, justice, trust, accountability, and respect are essential components of academic and educational integrity (Holden et al., 2021). The activity of doing fair and coherent study and finishing academic work is known as academic honesty (Guerrero-Dib et al., 2020; Ramdani, 2018). AI use in education is a hot topic right now, and there are a lot of benefits, such as cooperation, intellectual engagement, valuation, and accessibility. However, this development has also sparked questions concerning plagiarism and academic integrity (Cotton et al., 2023), which makes it possible for the standards of educational integrity to be violated (Roe & Perkins, 2022).

For example, when students utilise chatbot applications, it can be difficult for academics to evaluate their comprehension of the materials because it could not properly depict their actual educational background (Cotton et al., 2023). In a similar vein, the use of AI by students opens up new avenues for technologically advanced forms of academic dishonesty and norm and standard violations, such as research plagiarism, improper learning methods, and transgressions (Roe & Perkins, 2022). The idea is that "students' integrity hangs in the scale" as ChatGPT usage is a two-edged sword that can be used to commit academic fraud and plagiarism or to reduce effort, improve task performance and usefulness, increase academic self-efficacy, and lessen anxiety. In light of this, it is maintained that gathering insightful data requires looking at the moderating effect of students' integrity when utilising ChatGPT. While integrity has been extensively documented in academic literature and demonstrated to have a direct influence on behaviour (Cerdà-Navarro et al., 2022; Cotton et al., 2023; Holden et al., 2021) as well as being a valid moderator variable in academic contexts (Bin-Nashwan, Ismaiel, et al., 2023), very few studies had examined its moderator role in strengthening the links between numerous related constructs, particularly with regards to students' ChatGPT utilisation. As such, this study formulates the following hypotheses:

H7. Students' integrity forecasts use of ChatGPT.

H7a. Students' integrity moderates positively the relationship between performance expectancy and the usage of ChatGPT.

H7b. Students' integrity moderates positively the relationship between the effort expectancy and the usage of ChatGPT.

H7c. Students' integrity moderates positively the relationship between social influence and the usage of ChatGPT.

H7d. Students' integrity moderates positively the relationship between educational self-efficacy and the usage of ChatGPT.

H7e. Students' integrity moderates positively the relationship between technology self-efficacy and the usage of ChatGPT.

H7f. Students' integrity moderates positively the relationship between personal anxiety and the usage of ChatGPT.

3. Research methodology

3.1. Design, sampling and data collection procedure

Since quantitative research is suited for analysing causal models and linkages, it was used in this study. It addresses theory-based research questions and hypotheses by carefully manipulating a small number of variables, with the ultimate goal of offering a thorough summary of trends and correlations (Sekaran & Bougie, 2019). The study used a web-based survey in conjunction with a cross-sectional research approach to validate the suggested ChatGPT usage model. The electronic survey technique was chosen because of its attractive attributes, including cost-effectiveness, responsiveness, and accessibility to the target population (Sekaran & Bougie, 2019; Wright, 2006), and since the current study is focused on a technology-related topic. Students at universities make up the target sample for this study and deemed reliable sample (Febrilia et al., 2024). The researchers shared the survey URL with their academic peers across neighbouring ASEAN member states, including Malaysia, Indonesia, Singapore, Brunei, Thailand, and the Philippines. Subsequently, these colleagues randomly distributed the survey to students from non-science majors of different years for dissemination. For data analysis, researchers obtained 921 valid responses, achieving a response rate of 61.4%, which is considered satisfactory and higher than the typical response rate for online surveys, as noted by Wu et al. (2022).

3.2. Measurements and analysis tool

A total of 41 measurement items comprise the online survey used in this study to assess the eight research model constructs. To ensure the validity of the instruments and their fit to the research setting, measuring scales from previous studies were modified as deemed possible. A five-point Likert scale with anchors from “strongly disagree” (1) to “strongly agree” (5) was employed to measure the multi-item variables. Table 1 enumerates the items together with their sources. Before distributing the questionnaire, the researchers obtained three academic specialists in quantitative research to evaluate it. After that, 30 students participated in a pilot test of the questionnaire, whereby the results were satisfactory.

4. Report of data analysis

4.1. Sample description

Missing values were assessed by performing descriptive statistics analysis utilising IBM-SPSS 28; however, the results revealed none. Meanwhile, multivariate outliers were detected using the standard method of calculating the squared Mahala-Nobis distance at $p < 0.001$ for every case in the dataset. A total of 22 out of the 943 cases were determined as multivariate outliers, leading to their omission (Tabachnick & Fidell, 2013). Hence, 921 cases were valid for data analysis. Table 2 summarises the demographic statistics for the sample.

4.2. Common method bias

Considering the single-source data collection, we evaluated common method bias (CMB) to guarantee the accuracy of the study findings. Therefore, in order to ascertain if a single construct can explain a meaningful portion of the variance in the model, this present work employed Harman’s single-factor analysis. The findings indicate that the greatest single construct makes up 29.17% of the variance. This amount is less than Podsakoff et al.’s (2003) advised upper limit of 50%. Thus, the validity of CMB in the model is not an issue.

In addition, the study carried out a thorough analysis of collinearity by applying Kock’s (2015) test of full collinearity to ascertain whether the variance inflation factor (VIF) values exceeded the upper limit of 3.3.

Table 1

Construct operational definition, measurement items, and sources.

Constructs	Description	No Items	Source
Use of ChatGPT	The usage of ChatGPT.	5	(Papacharissi & Rubin, 2000; Venkatesh et al., 2003)
Perform Expectancy	ChatGPT is beneficial and improves task performance.	4	Venkatesh et al. (2003)
Effort Expectancy	Convenience and usability of using ChatGPT.	4	Venkatesh et al. (2003)
Social influence	The opinions of certain people affect the individual’s actions in relation to ChatGPT usage.	4	Venkatesh et al. (2003)
Educational self-efficacy	Confidence in achieving success and educational goals.	4	(Carol Midgley et al., 2000)
Technology self-efficacy	Users’ perception of their capabilities to utilise ChatGPT to conduct tasks and achieve intended outcomes.	5	Compeau and Higgins (1995)
Personal Anxiety	An individual’s level of trepidation linked to the projected or actual usage of ChatGPT upon interacting with the technology.	4	Rana et al. (2017)
Student Integrity	The habit of investigating and finishing educational tasks in a fair, honest, and coherent manner.	11	Ramdani (2018)

This study employed the partial least squares structural equation modelling (PLS-SEM) in Smart PLS 4 developed by Ringle et al. (2022). PLS-SEM is a reliable technique for looking at complex interactions between latent constructs. Because of its ability to handle non-normal data distributions and intricate interactions between latent components, PLS-SEM has grown in prominence in the social sciences and management domains (Hair et al., 2019). PLS-SEM is especially useful for investigations that seek to predict results instead of establishing causal links across variables, in contrast to the covariance-based SEM (CB-SEM). This is on top of its suitability during the investigation phase for constructing theories and forecasting or expanding upon an already-existing structural theory.

Table 2

Demographic information of the Study’s sample.

Variable	Category	Frequency	Percentage
Gender	Female	589	64%
	Male	332	36%
Age	18–30 years	617	67%
	31–40 years	276	30%
	40 years and above	28	3%
Study Program	First Degree - Bachelor’s Degree	443	48%
	Postgraduate - Master’s Degree	295	32%
	Postgraduate – PhD, DBA Degree	110	12%
	Others, e.g., Professional Certificate	73	8%
Study Mode	Course Work	598	65%
	By Research	222	24%
	Mix Mode (Course Work + Research)	101	11%
Study Time	Full-Time	801	87%
	Part-Time	120	13%

In this case, every variable was regressed against a common variable; the absence of single-source bias is indicated by a VIF ≤ 5 . VIF values of 3.3 and below were yielded from the analysis, the model can be considered free of common method bias. (see Table 3).

4.3. Measurement model assessment

It is crucial to guarantee the validity and reliability of the variables to avoid errors in the research findings and to make appropriate conclusions. In this study, the validity and reliability of the variables were assessed as part of the reflective model evaluation. It is essential to

Table 3
Full collinearity.

Variables	VIF
Use of ChatGPT	3.187
Perform Expectancy	3.155
Effort Expectancy	2.565
Social influence	1.759
Educational self-efficacy	2.933
Technology self-efficacy	2.718
Personal Anxiety	3.104
Student Integrity	2.460

establish criteria for dependability, convergent validity, discriminant validity, and internal consistency (Hair et al., 2019). Table 4 illustrates the retained item loadings for the model's assessment; they are shown to be substantially accepted as the items have achieved the threshold values for factor loading (≤ 0.5), AVE (≤ 0.5), and CR (≤ 0.7). Two items (Inti10 and Inti11) showed poor loading of less than 0.5; hence, they were removed as per the suggestion of Hair et al. (2018).

Discriminant validity was employed to determine the HTMT ratios (Kline, 2016). Table 5 shows that all the constructs have HTMT values of 0.85 and lower (Henseler et al., 2015), indicating the non-presence of discriminant validity issues. Therefore, the validity of the measurement model can be concluded, suggesting adequate differences between the constructs. Based on the statistical tests performed on the measurement model, it was confirmed that the variables' measurement scales were valid and reliable, thus enabling the assessment of the structural model.

Table 4
Validity and reliability results.

Constructs	Indicators	Loadings	CR	AVE
Use of ChatGPT	ChatGPT1	0.895	0.913	0.741
	ChatGPT2	0.866		
	ChatGPT3	0.922		
	ChatGPT4	0.889		
	ChatGPT5	0.717		
Performance Expectancy	PE1	0.820	0.864	0.701
	PE2	0.906		
	PE3	0.819		
	PE4	0.802		
Effort Expectancy	EE1	0.883	0.932	0.691
	EE2	0.792		
	EE3	0.776		
	EE4	0.869		
Social Influence	SI1	0.816	0.848	0.689
	SI2	0.896		
	SI3	0.884		
	SI4	0.709		
Educational self-efficacy	ASE1	0.860	0.844	0.678
	ASE2	0.736		
	ASE3	0.819		
	ASE4	0.872		
Technology self-efficacy	TSE1	0.856	0.911	0.719
	TSE 2	0.787		
	TSE 3	0.896		
	TSE 4	0.894		
	TSE 5	0.800		
Personal Anxiety	PA1	0.743	0.849	0.674
	PA2	0.907		
	PA3	0.817		
	PA4	0.809		
Students' Integrity	Inti1	0.769	0.935	0.621
	Inti2	0.843		
	Inti3	0.780		
	Inti4	0.780		
	Inti5	0.727		
	Inti6	0.832		
	Inti7	0.811		
	Inti8	0.780		
	Inti9	0.765		

Note: Items Inti10 and Inti11 have been removed due to poor loading. All factor loadings are significant at $p < 0.05$.

4.4. Structural model assessment

This assessment of the structural model was performed to test all the hypotheses. Bootstrapping was employed using 5000 iterations following Hair et al. (2021). Table 6 presents the hypothesis testing results on the direct and moderated relationships. Strong positive relationships were indicated between some of the constructs, namely performance expectancy, social influence, educational and technology self-efficacy with ChatGPT usage. In contrast, integrity was revealed to be negatively correlated to ChatGPT usage in academia. As such, these outcomes support H1, H3, H4, H5 and H7, but reject H2 and H6. In terms of the moderated relationships, integrity was shown to significantly and positively moderate the link between effort expectancy and ChatGPT usage. In contrast, integrity showed a significantly negative moderating effect on the relationships between performance expectancy, technology self-efficacy, and ChatGPT usage. Meanwhile, the rest of the relationships were not affected by the moderator variable. Therefore, H7b is supported whilst H7a, H7c, H7d H7e and H7f are rejected. The explanatory power of the developed research model indicated a determination coefficient $R^2 = 82.4\%$ of the variance in ChatGPT usage, suggesting the model's has predictive accuracy, predominantly meeting the set criteria by various researchers (Hair et al., 2019, 2021; Shmueli et al., 2019).

Also examined was the PLS predict to ensure the validity of predictive relevance. The case-level predictions were calculated whereby $k = 10$, as per Hair et al. (2019) and Shmueli et al. (2019). The results demonstrate that the Q^2 predict values are beyond zero (Q^2 predict > 0), demonstrating the outperformance of the Linear Model (LM) benchmark by all the indicators. Hence, a comparative analysis between the RMSE values and the naïve LM benchmark could be made. The findings demonstrated that all the indicators produced larger prediction errors than the LM (PLS-SEM $<$ LM) (see Table 7).

5. Discussion and conclusions

5.1. Discussion

In this research, an integrated model of UTAUT-SCT provides a comprehensive framework to understand the multifaceted-dynamics shaping the adoption behaviour of ChatGPT among students. Our model contains the direct effects observed and the intriguing-results regarding the moderating effects of students' integrity on their adoption behaviour context.

In the direct relationships, we obtained some interesting outcomes matching the proposed UTAUT-SCT model. Students' performance expectancy and social influence emerge as robust drivers, strongly influencing their ChatGPT adoption. Performance expectancy-adoption relationship exerted a positive trend, aligning with UTAUT focus on technologies' perceived usefulness and the extant literature (Bin-Nashwan, 2022, 2023; Menon & Shilpa, 2023; Rahim et al., 2022; Strzelecki & ElArabawy, 2024). Within the AI-driven chatbot adoption perspective, students tend to view it as a valued tool for AI-based academic support or assistance in their assignments and studies. This innovative platform holds the capability to foster their learning experiences, aid in assignments, or facilitate understanding, which may contribute to its perceived usefulness, driving adoption behaviour.

Likewise, social factors-technology adoption relationship had a significant and positive, resonating with the theoretical frameworks' assumption in this context (Bouteraa et al., 2021; Ma and Huo, 2023; Ong et al., 2023; Strzelecki & ElArabawy, 2024; Tewari et al., 2023). The UTAUT-SCT model highlights the role of peers and social networks in shaping human behaviours. Within this framework, the influence of classmates, friends, or influential individuals who endorse AI-driven chatbots as useful tools may strongly sway students' decisions to adopt them. However, the study's results diverge from the UTAUT model by demonstrating a lack of significant effort expectancy-adoption

Table 5
Heterotrait-monotrait ratio (HTMT).

Constructs	1	2	3	4	5	6	7	8
1 Education self-efficacy								
2 Effort Expectancy	0.401							
3 Performance Expectancy	0.504	0.699						
4 Personal Anxiety	0.532	0.651	0.709					
5 Social Influence	0.390	0.529	0.497	0.583				
6 Technology self-efficacy	0.799	0.535	0.621	0.655	0.388			
7 Use of ChatGPT	0.640	0.598	0.734	0.674	0.524	0.713		
8 Student Integrity	0.542	0.591	0.699	0.668	0.683	0.553	0.583	

All values of HTMT ≤ 0.85 .

Table 6
Hypotheses testing.

Hypothesis	Relationship	Std. Beta	Std. Dev	t-value	p-value	PCI LL	PCI UL	f ²	Decision
H1	Perform Expectancy → Use ChatGPT	0.553	0.026	21.393	0.000	0.497	0.600	0.584	Supported
H2	Effort Expectancy → ChatGPT	0.025	0.023	1.103	0.270	−0.020	0.071	0.002	Not Supported
H3	Social influence → ChatGPT	0.108	0.018	5.851	0.000	0.074	0.145	0.039	Supported
H4	Educational self-efficacy → Use of ChatGPT	0.114	0.022	5.294	0.000	0.072	0.156	0.030	Supported
H5	Technology self-efficacy → Use of ChatGPT	0.229	0.028	8.059	0.000	0.181	0.293	0.039	Supported
H6	Personal Anxiety → Use of ChatGPT	0.144	0.026	5.637	0.000	0.094	0.193	0.038	Supported
H7	Student Integrity → Use of ChatGPT	−0.172	0.036	4.740	0.000	−0.245	−0.102	0.047	Supported
H7a	Perform Expectancy × Student Integrity → Use of ChatGPT	−0.076	0.028	2.753	0.006	−0.126	−0.017	0.012	Not Supported
H7b	Effort Expectancy × Student Integrity → Use of ChatGPT	0.144	0.031	4.709	0.000	0.086	0.208	0.051	Supported
H7c	Social influence × Student Integrity → Use of ChatGPT	−0.022	0.036	0.610	0.542	−0.106	0.034	0.001	Not Supported
H7d	Educational self-efficacy × Student Integrity → Use of ChatGPT	0.021	0.027	0.774	0.439	−0.027	0.080	0.001	Not Supported
H7e	Technology self-efficacy × Student Integrity → Use of ChatGPT	−0.182	0.034	5.348	0.000	−0.264	−0.133	0.098	Not Supported
H7f	Personal Anxiety × Student Integrity → Use of ChatGPT	−0.039	0.045	0.875	0.382	−0.110	0.064	0.003	Not Supported

Table 7
Q² predict and PLS predict performance.

Indicators	Q ² predict	PLS-SEM_RMSE	LM_RMSE
ChatGPT1	0.570	0.536	0.637
ChatGPT2	0.501	0.504	0.626
ChatGPT3	0.651	0.380	0.548
ChatGPT4	0.671	0.399	0.519
ChatGPT5	0.528	0.555	0.609

relationship, which raises intriguing questions. This result could be due to the specific characteristics of the student sample, such as prior exposure to similar technologies or individual preferences, which may affect how they perceive the ease of using ChatGPT. Different backgrounds and experiences may result in varied perceptions of effort-expectancy. The lack of support for the relationship between effort expectancy and ChatGPT usage also suggests that students' perceptions of the ease of using ChatGPT do not significantly influence their decision to adopt the technology. This finding may indicate that students do not perceive ChatGPT as particularly difficult or cumbersome to use, or that other factors play a more significant role in shaping their adoption behavior.

Moreover, educational self-efficacy-ChatGPT adoption relationship was revealed as significant. We can argue that students with greater degrees of educational self-efficacy are likely to believe in their ability to succeed academically. When these students encounter AI-driven chatbots, they may perceive them to boost their academic performance. This self-efficacy-adoption positive relationship demonstrates that students who are confident in their academic capabilities tend to adopt ChatGPT as an additional-resource to aid their learning, thus reinforcing their favourable attitudes toward its adoption (Cotton et al., 2023; Strzelecki, 2023; Tlili et al., 2023). Similar to educational self-efficacy, higher-levels of students' technology self-efficacy reflect their confidence in using technology effectively. Thus, students with greater technological self-efficacy are likely to view AI-driven chatbots as tools they can easily adapt to and utilise for academic purposes. Their confidence in using technology could positively shape their ChatGPT adoption, as they may

feel more comfortable and able to engage with such a tool effectively. In this model, we also found personal anxiety-ChatGPT adoption relationship was significantly positive, indicating that anxiety about academic performance could stimulate students to seek tools that alleviate stress or improve their capabilities (Nguyen & Chen, 2023). AI-driven chatbots like ChatGPT, perceived as effective instruments that aid in understanding, completing assignments, or minimising academic-workload, may attract students who are experiencing high-levels of anxiety. These individuals may see these innovative tools mitigate their anxiety and enhance their academic outcomes, thus increasing their intention to adopt them.

The observed negative significant relationship between student integrity and the utilisation of ChatGPT within higher education settings signals a noteworthy trend wherein integrity emerges as a predictive factor for ChatGPT adoption among students. This finding implies that students' ethical standards and commitment to academic integrity play a decisive role in influencing their inclination towards integrating ChatGPT into their academic pursuits. With higher levels of integrity correlating with diminished ChatGPT usage, it suggests that individuals with strong ethical principles may exhibit more cautious or restrained attitudes towards incorporating ChatGPT technology into their academic practices. This discernment underscores the intricate interplay between ethical considerations and technological assimilation within higher education, underscoring the imperative of nurturing a culture of integrity to foster responsible and ethical utilisation of AI technologies such as ChatGPT. Conversely, students with lower levels of integrity may be more inclined to use ChatGPT as a means of gaining unfair advantage or circumventing academic standards.

Interestingly, we found that students' integrity-moderated interaction of effort expectancy and ChatGPT adoption is significant. It implies that the effort expectancy-ChatGPT adoption behaviour relationship varies based on the different levels of integrity that the students possess. Higher-levels of integrity significantly enhance the relationship. The results indicate that the direct effect of effort expectancy on adoption was altered to be significant with the presence of students' ethical considerations. However, students' integrity-moderated interaction of

performance expectancy and ChatGPT adoption was insignificant; this might be a result of the perceived usefulness nature. Although a significantly direct performance expectancy-adoption relationship, students' integrity-moderated interaction was marginal. Usefulness of AI-driven chatbots including ChatGPT among students remain consistent despite their integrity levels, demonstrating integrity-moderated interaction exerted no effect.

Likewise, integrity-moderated interaction of social influence and ChatGPT adoption was not empirically significant, suggesting that peer influence may function independently of ethical considerations. Social influence likely has a robust role in adoption decisions regardless of integrity-levels, signifying that ethical concerns may not significantly alter the influence of peers or social networks. Further, the study showed no significant integrity-moderated interaction between self-efficacy and ChatGPT adoption; this might be owing to the more internally-focused nature of these beliefs. We also found that educational self-efficacy mostly concerns individuals' confidence in their abilities rather than ethical considerations, as this factor is likely more self-referential. However, although there was a positive direct relationship, the presence of integrity changed the technology self-efficacy-adoption relationship to be significantly negative. This could be attributed to the fact that despite perceived confidence in AI-driven chatbots among students, they may have some concerns about ethical considerations in relation to their AI adoption and fear of over-reliance, weakening their adoption behaviour. Furthermore, the work revealed that integrity showed no significant integrity-moderated interaction between personal anxiety and adoption. Additionally, personal anxiety might drive students to seek solutions without being strongly moderated by integrity-related ethical concerns.

5.2. Theoretical implications

This study represents a significant theoretical contribution to the understanding of students' adoption behavior regarding OpenAI, particularly ChatGPT, within educational contexts. By integrating the Unified Theory of Acceptance and Use of Technology (UTAUT) and Social Cognitive Theory (SCT), the research offers a nuanced perspective on the factors influencing students' decision-making processes when engaging with AI-driven educational tools. The integration of UTAUT and SCT provides a comprehensive framework for analyzing user behavior. By combining these theories, the study acknowledges the multifaceted nature of technology adoption and highlights the interplay between individual beliefs, social dynamics, and environmental factors.

Performance expectancy and effort expectancy, derived from UTAUT, assess users' perceptions of the usefulness and ease of use of ChatGPT, respectively. These factors reflect users' cognitive assessments of the benefits and costs associated with adopting the technology. Additionally, social influence, educational self-efficacy, technology self-efficacy, and personal anxiety, informed by SCT, capture the social and psychological dimensions of adoption behavior. These variables recognize the influence of peers, the confidence in one's abilities to utilise the technology effectively, and the presence of apprehensions or concerns about its use. Furthermore, the study's emphasis on empirical validation fills a crucial gap in the existing literature on AI adoption in educational settings. By empirically testing the proposed model, the research provides empirical evidence supporting the theoretical framework and enhances its practical utility. This validation contributes to the establishment of a robust foundation for understanding and predicting students' behavior towards AI-powered educational tools.

The inclusion of students' integrity as a moderating factor in the study of ChatGPT adoption represents a significant theoretical advancement, particularly in addressing the ethical dimensions of AI-driven educational tools within academia and higher education. Firstly, integrating integrity as a moderating factor expands the theoretical framework by acknowledging the complex interplay between technological adoption and ethical considerations. While existing

models such as UTAUT and SCT focus primarily on cognitive, social, and psychological factors influencing adoption behavior, the incorporation of integrity highlights the importance of moral principles and values in technology use. This addition enriches the theoretical understanding by recognizing the ethical implications inherent in the deployment of AI chatbots in educational settings.

Moreover, the emphasis on integrity underscores the broader societal discourse surrounding the ethical use of AI technologies, especially in educational contexts. By considering students' integrity as a moderating factor, the study prompts critical reflection on the ethical responsibilities of both users and developers of AI-driven tools. It encourages stakeholders to consider questions of transparency, accountability, and fairness in the design and implementation of such technologies, thus contributing to the larger conversation about the ethical implications of technology in learning environments. Hence, by addressing the ethical implications and potential threats of AI chatbots, the research enriches the broader conversation about the responsible integration of technology in learning environments and underscores the importance of ethical considerations in shaping technological adoption behavior.

5.3. Practical implications

The research outcomes yield crucial practical implications stemming from the associations between the utilisation of ChatGPT or other AI-driven educational tools in higher education, along with various influencing factors, including student integrity as a potential moderator, hold significant relevance for research and application within higher education environments. Understanding these relationships offers valuable insights for designing interventions and strategies aimed at fostering the effective and ethical integration of ChatGPT technology in educational settings.

- **Perform Expectancy:** The positive relationship between perform expectancy and the use of ChatGPT suggests that students' belief in their ability to effectively utilise the technology influences its adoption. In practical terms, educational institutions can focus on bolstering students' confidence in their capacity to use ChatGPT by providing training programs, workshops, or tutorials. Additionally, highlighting success stories and showcasing the benefits of ChatGPT in educational contexts can further enhance perform expectancy and encourage its usage.
- **Effort Expectancy:** Despite not finding significant support, the notion that effort expectancy does not directly influence the use of ChatGPT indicates the need for further investigation into why students may perceive using ChatGPT as requiring substantial effort. Practical implications may include redesigning user interfaces to enhance usability, streamlining processes, and providing support resources to mitigate perceived effort barriers associated with ChatGPT usage. Designing programs that are directed at training and enhancing students' educational and technological self-efficacy is a point that deserves attention from relevant stakeholders. This can be done by offering workshops or resources that bolster students' confidence in adopting technology effectively and persuade them to harness the full potential of AI-platforms for academic enhancement.
- **Social Influence:** The positive relationship between social influence and ChatGPT usage underscores the importance of peer recommendations, instructor endorsements, and social networks in facilitating technology adoption. Institutions can capitalize on this by fostering a culture of technology adoption through peer mentoring programs, incorporating ChatGPT into collaborative learning activities, and leveraging influencers within the educational community to promote its use.
- **Educational Self-Efficacy:** The finding that educational self-efficacy positively influences ChatGPT usage highlights the significance of students' confidence in their academic abilities. Practical

implications involve implementing strategies to enhance academic self-efficacy, such as providing personalized feedback, setting achievable goals, and offering support mechanisms tailored to individual learning needs.

- **Technology Self-Efficacy:** Similar to educational self-efficacy, technology self-efficacy plays a crucial role in shaping students' readiness to adopt ChatGPT. Institutions can support technology self-efficacy by offering technology training programs, fostering a culture of experimentation and innovation, and providing technical support to address any challenges students encounter in using ChatGPT.
- **Personal Anxiety:** The negative relationship between personal anxiety and ChatGPT usage suggests that anxiety can act as a barrier to technology adoption in educational settings. To mitigate this, institutions should prioritize creating a supportive learning environment that addresses students' anxiety through counseling services, stress management techniques, and promoting a growth mindset towards technology adoption.
- **Student Integrity:** The lack of support for moderating effects by student integrity suggests that integrity considerations do not significantly alter the relationships between independent variables and ChatGPT usage. However, it highlights the importance of fostering a culture of integrity and ethical behavior in technology adoption processes. Practical implications involve integrating discussions on academic integrity and responsible technology use into curriculum, establishing clear guidelines and policies regarding ChatGPT usage, and promoting ethical decision-making among students.

5.4. Research limitations and future directions

Despite the meaningful discoveries that emerged from this empirical attempt, some limitations were identified. Given that this research predominantly utilised a positivist paradigm by employing the cross-sectional quantitative method, further constructivist and/or pragmatic paradigms through mixed method investigations are encouraged in the future by merging the quantitative and qualitative approaches to investigate more nuanced aspects to advance a triangulated conclusion.

In addition, while the theoretical model is robust and well established, it may be beneficial to replicate the study or conduct additional tests with diverse samples and across various settings to validate the present research findings (Chekima & Chekima, 2019; Chekima et al., 2023; Juliana et al., 2022).

Employing bibliometric analysis is a further suggestion for enhancing the domain of generative AI. It adds credibility to systematic literature reviews and meta-analyses and serves as a robust method for examining and interpreting extensive scientific data. This approach allows researchers to delve into the subtle shifts in a specific field's development and shed light on its emerging areas. Bibliometric analysis can yield significant and interconnected insights into generative AI tools in education, aiding in understanding the broader intellectual landscape (Nicole et al., 2022; Lada, Chekima, Ansar, et al., 2023; Lada et al., 2023b). However, its utilisation in business research remains relatively novel and frequently inadequate, particularly concerning the acceptance and adoption behavior of generative AI tools.

ChatGPT is the main focus of this study owing to its popularity among the academic community. However, it did not include other types of AI language model platforms. Future studies might investigate different forms of Chatbots that are equally evolving with specific advantageous features than ChatGPT, which can assist academics in completing various tasks.

Furthermore, introducing demographic factors such as education level, study program, and mode as the moderators in the ChatGPT model would contribute to greater comprehension regarding the behaviour of students through multigroup analysis. In terms of the target population, future studies may focus on the policymakers and staff of academic institutions to obtain their views on AI language model adoption.

CRedit authorship contribution statement

Mohamed Bouteraa: Writing – review & editing, Writing – original draft, Project administration, Data curation, Conceptualization. **Saeed Awadh Bin-Nashwan:** Writing – review & editing, Writing – original draft, Conceptualization. **Meshari Al-Daihani:** Writing – original draft, Resources. **Khadar Ahmed Dirie:** Writing – original draft, Conceptualization. **Abderrahim Benlahcene:** Writing – review & editing, Methodology. **Mouad Sadallah:** Writing – review & editing, Validation, Formal analysis. **Hafizah Omar Zaki:** Writing – review & editing, Software, Funding acquisition. **Suddin Lada:** Writing – review & editing, Funding acquisition, Conceptualization. **Rudy Ansar:** Funding acquisition, Investigation. **Lim Ming Fook:** Validation, Visualization. **Brahim Chekima:** Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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